

Manual

1. About the Program:

The program focuses in the simulation, forecasting, parameters determination, and visual representation of vector born diseases. Different vector born diseases can be represented depending on the parameters used in the pre-configuration steps. Furthermore, the program handles commuting traveling and weather conditions as driver in the simulation and forecasting.

The commuting is handled as a relative percentage of the time a given population spent on a **Cluster**. a Cluster is a region or zone that could be specified by the user (by latitude and longitude coordinates) or as a general region with a given population. The Number of clusters imply how many regions or zones are considered in the forecasting or simulation process.

The weather conditions are considered that affect the mosquito life cycle (e.g. high temperatures and low temperatures can produce high mortality rates on a given mosquito population). Weather variables are considered to NOT produce any effect on host populations. The program can handle up to three environmental variables at the same time (e.g. relative humidity, temperature, vegetation index, etc.). Considerable care must be taken in the parameters used since if their values are considerable biased to a given central point (e.g. 25 °C) the models can produce error or stop estimating given that exponential expressions are used to handle environmental variables. This implies that big differences (by an improper weather variable handling) can lead to huge numbers that could lead to computational errors.

The software is based on the classical SEIR-SEI model with modifications. In this manual first it is covered the theoretical concepts regarding the models. Later the software is explained together with its use.

2. About the Models:

Mechanistic models have extensively used in conjunction with the compartmental methodology (i.e. divide the total population into "compartments") in order to represent the dynamic behaviours of the host and vectors at different stages of the epidemic states. The widely adopted SEIR-SEI model is an ordinary differential equation generalization of the Ross-MacDonald model¹. The SEIR-SEI model focuses in the representation of the vector N_v and human N_h dynamics. The allowed states (compartments) are defined depending on the disease stages; susceptible (free of dengue virus; S), exposed (carrying the dengue virus but incapable to transmit it yet; E), infectious (carrying the dengue virus and ready to transmit it, I), and recovered (the host component can describe a disease recovered state while vectors are not considered to be recovered; R). The transition between the compartments depend on the complexity involved in the representations that could consider (but not limited) to: multi-serotypes, mosquitoes movement (diffusion), host traveling tendencies, dengue fever state, sub-models for host and vectors, networks between regions, among other disease behaviours representations.²

¹ Smith et al., "Ross, Macdonald, and a Theory for the Dynamics and Control of Mosquito-Transmitted Pathogens."

² Esteva and Vargas, "Analysis of a Dengue Disease Transmission Model."

The mosquito life-cycle is generally ignored since the different pre-adult stages (egg, larvae, pupae, and immature mosquito) are not involved in the disease transmissions. This simplifications can be considered useful for model simplifications, nevertheless there are considerable factors, such as environmental drivers (average temperature, minimum temperature, maximum temperature, humidity, vegetation, rainfall, among others), that are required to predict an unbalance in the mosquitoes populations that could lead to diseases outbreaks. Different works have made some approaches to incorporate different environmental drivers in their models³. The classical vector models are first described in this section followed by description of several modifications implemented in this software in order to extend the models applicability.

In the SEI model is assume that **total population, N_v** , is subdivided into three independent but interconnected compartments (Susceptible, Exposed, and Infected). The corresponding compartment population are denoted by **$S_v(t)$, $E_v(t)$** , and **$I_v(t)$** , respectively. Equations 1 to 3 describe the classical compartment model. The main interaction with hosts is given by the last term of Equation 1, which depend of the per-capita contact rate. The per-capita contact rates for the vector and host are shown in Equations 10 and 11, respectively.

$$\frac{dS_v(t)}{dt} = \mu_v N_v - \mu_v S_v(t) - \beta_v \frac{I_h(t)}{N_h} S_v(t) \quad (1)$$

$$\frac{dE_v(t)}{dt} = \beta_v \frac{I_h(t)}{N_h} S_v(t) - (\mu_v + a_v) E_v(t) \quad (2)$$

$$\frac{dI_v(t)}{dt} = a_v E_v(t) - (\gamma_v + \mu_v) I_v(t) \quad (3)$$

As made in different works, it can be assume that the population of mosquitoes is constant and therefore **$S_v(t) + E_v(t) + I_v(t) = N_v$** . This is normally made constant⁴ by the assumption that that birth and death rates are equal; in general the total population of vector is a variable (**$N_v(t)$**) affected by different drivers (e.g. environment, mobility, disease, control measures, etc.). In this software, the total vector population was considered a variable.

The classical human model correspond to the SEIR component of the global model. The total host population, **N_h** , is subdivided into four classes (susceptible, exposed, infectious, and recovered). The corresponding population sizes of these four classes are denoted by **$S_h(t)$, $E_h(t)$, $I_h(t)$** , and **$R_h(t)$** , respectively. An additional variable can also be used **$C_h(t)$** in order to represent the accumulated population of infected host. This variable can be used for models parametrization. The set of differential equations dedicated to the host compartments are specified from Equation 4 up to 8. The last term of Equation 4 describe correspond to the human-vector interaction.

Since the host population is almost invariant during a short modelling process, it can be assumed without mayor error that **$S_h(t) + E_h(t) + I_h(t) + R_h(t) = N_h$** . The per-capita contact rates values can be obtained by using the definition of the Ross-Macdonald models. The host contact rate depends on the mosquito biting rate (α), the probability of the pathogen to be transmitted from the vector to the host, and the ratio of population between the vector to the human in the given region. Similarly, the

³ Hoshen and Morse, "A Weather-Driven Model of Malaria Transmission"; Mordecai et al., "Optimal Temperature for Malaria Transmission Is Dramatically Lower than Previously Predicted"; Beck-Johnson et al., "The Effect of Temperature on Anopheles Mosquito Population Dynamics and the Potential for Malaria Transmission."

⁴ Barrios, Lee, and Vasilieva, "Assessing the Effects of Daily Commuting in Two-Patch Dengue Dynamics."

vector contact rate depends on the biting rate and the pathogen transmission in the opposite direction, but not on the ratio of host to vectors, (Eq. 10 and 11) ⁵.

$$\frac{dS_h}{dt} = \mu_h N_h - \mu_h S_h(t) - \beta_h \frac{I_h(t)}{N_h} S_h(t) \quad (4)$$

$$\frac{dE_h}{dt} = \beta_h \frac{I_h(t)}{N_h} S_h(t) - (\mu_h + k_h) E_h(t) \quad (5)$$

$$\frac{dI_h}{dt} = k_h E_h(t) - (\gamma_h + \mu_h) I_h(t) \quad (6)$$

$$\frac{dR_h}{dt} = \gamma_h I_h(t) - \mu_h R_h(t). \quad (7)$$

$$\frac{dC_h}{dt} = k_h E_h(t) \quad (8)$$

$$\beta_v = \alpha P_v \quad (10)$$

$$\beta_h = \alpha P_h \frac{N_v}{N_h} \quad (11)$$

Modification of the classical vector model (SEI) was performed by connecting the different mosquito life stages with the classical SEI model. Specifically, Mosquitoes have four main development stages (egg, larva, pupae, and mosquito). Since only female mosquitoes can lay eggs, the development stages have to take into account that limitations. Furthermore, since the case studied is based on Dengue virus, as referenced, only females can bite humans. This consideration can lead to a double set of differential equations (or clusters), one represent males and another representing females, or considering that specific parameters obtained during model parametrization are only representatives of the female mosquitoes. The last alternative was considered in this software. Mosquito stage was also be divided in immature and adult stages. One of the important difference is that the immature mosquito do not bite and therefore is not counted for the transmission disease. The extrinsic incubation period (pathogen's latent period in the mosquito) is considered to be relatively long once compared to the mosquito life cycle. Therefore age structures, and suitable representations of each age structure could allow better model representability. The importance of age structure considerations is specially detrimental if it is desirable to incorporated in specific stages control techniques. Furthermore, stage structures are also important if it is considered that environmental variables play different roles depending on the mosquito stages (e.g. mortalities and reproduction in function of weather conditions).

The set of equations describing the different mosquito stages are described from Equation 12 up to Equation 16. In these equation each of the five proposed states are considered (Egg, $G(t)$; Larvae, $L(t)$; Pupae, $P(t)$; immature mosquito, $S_{v,im}(t)$; mature mosquito, $S_{v,a}(t)$). As observed in each equation the changes on each stages depend on previous stage states, given by a suitable development rates (D_i), a mortality rates (μ_i), and its own development rate. Furthermore, an interspecific and intraspecific effect is observed at the larva stage that can limit the mosquito further development (i.e. competitive survival). This inter-intraspecific is described by the last term of Equation 13.

⁵ Barrios, Lee, and Vasilieva.

$$\frac{dG}{dt} = \epsilon D_{ag} S_{v,a} - \mu_g G - D_E G \quad (12)$$

$$\frac{dL}{dt} = D_E G - \mu_l L - D_l L - \kappa L^2 \quad (13)$$

$$\frac{dP}{dt} = D_l L - \mu_p P - D_p P \quad (14)$$

$$\frac{dS_{v,im}}{dt} = D_p P - \mu_{v,im} S_{v,im} - D_i S_{v,im} \quad (15)$$

$$\frac{dS_{v,a}}{dt} = D_i S_{v,im} - \mu_{v,a} S_{v,a} \quad (16)$$

In order to generate and incorporate a general expression for environmental drivers effect on the different parameters involved in previous stage representations, an Arrhenius-based expression with a parameter-dependent central point (here and after refereed as braking point) were considered. The general formulation can be seen in Equation 17 in which the positive or negative value of the parameter under consideration (e.g. here applied to the mortality rate of mosquitoes egg) give reference to the increasing or decreasing shape of the values once the considered environmental driver variables are moved away from the braking point (i.e. mortality rates increase, positive sign, while development rates decreases, negative sign, as they move away from the braking point). The exponential term x helps to shape the degree of effect of the driver. Higher the value steeper effect of drivers. By allowing its value a multiple of two, a symmetric behaviour is obtained. The population reduction in each mosquito stage were considered to be given by the corresponding mortality rates (μ) and the development rates (D).

$$\mu_g = \mu_{g,o} e^{\pm \left(\frac{T - T_o}{\Delta E} \right)^x} \quad (17)$$

Even though the previous expression could have been used in each parameter involved (including some host parameters), the number of parameters to be environmental driver dependent were limited in order to restrict the number of new parameters that will be included in the ordinary differential equation system. This was done to allow wider generalization of the system and to limit the parameters that will have to be specified during the model definition. Specifically, the parameters that were considered temperature dependent were: each vector mortality rate and the egg development rate (i.e. incubation of the eggs) ($\mu_g, \mu_l, \mu_p, \mu_{v,im}, \mu_{v,a}, D_{ag}$). Extension of equation 17 to include further environmental drivers was done by incorporation further exponential terms in a multiplicative manner (with the respective modification of the T, x, T_o and ΔE by corresponding environmental driver parameters). In fact, Temperature and rainfall are known to be the relevant drivers that can affect the mosquitoes dynamics), therefore a multiplicative term could be considered. Regarding vector mobility, it was not considered in this work. This assumption would not impose a major burden to the model since mosquito diffusion tend to be more restrictive than host demographical changes when a broad demographic sector is consider (e.g. movement within cities).

Previous equations were interconnected to the classical vector model by use of the ordinary differential equations describing the non-infected adult vector population (Equations 1 and 16). Specifically, the last term of Equation 1 was included in Equation 16 in order to represent the transition from susceptible adults to the expectant stages (18). Furthermore, the total vector population was considered a state variable that was dependent on the development stages and therefore dependent of environmental variables (i.e. $N_v = f(t, T)$).

The stage transition model do not consider in it the infectious state of adults. The inclusion of infected components can be translated into include each vector stage in the oviposition stage (Equation 12).

The modification of the host stages can be performed in different ways. They can include asymptomatic individuals, migration/commuting, multi-serotype, vaccination considerations, seasonality by the use of time varying parameters, among other modifications. In the software, the host population was modified by considering the possibility of reinfections by other serotype. Demographical modifications was also considered (i.e. commuting, immigration, and emigration). Reinfection was considered after a person become susceptible again after a span given by a specific rate (τ), which implied the modification of equations 4 and 7 to:

$$\begin{aligned}\frac{dS_h}{dt} &= \mu_h N_h - \mu_h S_h(t) - \beta_h \frac{I_h}{N_h} S_h + \tau_h R_h(t) \\ \frac{dR_h}{dt} &= \gamma_h I_h(t) - (\mu_h + \tau) R_h(t)\end{aligned}$$

Commuting is basically performed by the use of the commuting matrix (Q). This matrix represent how much time a person of the cluster i spent in the cluster j . Careful must be taken into the system configuration since ideally the sum per line in a q matrix should be made equal to 1. If values bigger than one are used, they should be standardized for further applications and properly manipulated for parameter determination. Considering the commuting, the set of ODEs used are (for the SEIR-SEI model, only the SEIR component):

- $dS_h/dt(i) = u_h(i) \cdot N_h(i) - S_h(i) \cdot \text{sumah}(i) - u_h(i) * S_h(i) + \tau(i) * R_h(i)$
- $dE_h/dt(i) = S_h(i) * \text{sumahost}(i) - (u_h(i) + k_h(i)) * E_h(i)$
- $dI_h/dt(i) = k_h(i) * E_h(i) - (\gamma(i) + u_h(i)) * I_h(i)$
- $dR_h/dt(i) = \gamma(i) * I_h(i) - (u_h(i) + \tau(i)) * R_h(i)$
- $dC_h/dt(i) = k_h(i) * E_h(i)$
- $\text{sumav}(i) = \sum_j P(j, i) * I_h(j) / N_h(j)$
- $\text{sumah}(i) = \sum_j q(i, j) * I_v(j) * \beta_h(j) / N_v(j)$
- $\beta_v(i) = \alpha(i) * p_v(i)$
- $\beta_h(i) = \alpha(i) * p_h(i) * N_v(i) / \text{div}(i)$
- $P(i, j) = q(i, j) * N_h(i) / \text{div}(j)$
- $\text{div}(i) = \sum_j (\sum_k q(j, i) * N_h(j)) \quad \forall i - \forall j$

3. Software:

Installation: The program is composed of the following matlab files (an explanation of what each files do).

- *IngresoDatos.m*: Matlab main program that access different functionalities and open other m files depending of the requirements. Since this is a guide, you also need the *IngresoDatos.fig* file to run.
- *Run.m*: Introductory windows that once opened gives the alternative to open the *IngresoDatos.m* file.
- *ODEs.m*: This file calls the different differential equations and optimizations required during the simulation, parameter prediction, and forecasting process. Specific simulations (i.e. kalman filter, particle filters, and ODE based). The full differential equations can be found in the ODEs operated by this m file and correspond to the following:
 - *dengueODE1.m*: This is th classical SEIR-SEI model.
 - *dengueODE3_Modified.m*: This is the LSEI model.
 - *dengueODE4_Modified.m*: This is the SEIR-LSEI model.
 - *dengueODE7_Modified.m*: This is the LSEI model used for optimization (parameter determination).

- dengueODE9.m: This is the classical SEIR-SEI model with the Extended-Kalman-Filter.
- dengueODE10_Modified.m: This is the LSEI model with the Extended-Kalman-Filter.
- dengueODE11_Modified.m: This is the SEIR-LSEI model with the Extended-Kalman-Filter.
- dengueODE12.m: This is the classical SEIR-SEI model with the Particle-Filter.
- dengueODE13_Modified.m: This is the LSEI model with the Particle-Filter.
- dengueODE14_Modified.m: This is the SEIR-LSEI model with the Particle-Filter.
- dengueODE15.m: This is the SEIR-SEI model for optimization (parameter determination)
- dengueODE16_Modified.m: This is the SEIR-LSEI model used for optimization.
- MeasurmentFcnDengue.m: Extended Kalman filter Measurement Function.
- vdpMeasurmentLikelihoodFcnEduardo.m: Likelihood function for the particle filter.
- Cholcoved: Cholesky-like covariance decomposition
- Further supporting files, exercises files, such as images and map are also included and therefore all the files included should be placed on a same file.

To open the software first, you have to run the IngresoDatos.m matlab file or the Run.m matlab file. The Run.m file opens an introductory screen to open the IngresoDatos.m file so both end at the same point. The following figure shows the opening User Interface opened throughout the matlab files. As observed in the figure, there are seven different tabs that the user can interact with the software. The “**File**” tab have the *open*, *close*, *save*, and *load* states options. The “**Load Data**” tab opens a new screen that allows loading the data in different formats directly from your computer or from the database. The screens opened depend on chooses made. Figure 2 shows the Load Data>> From Computer screen while Figure 3 shows the Load Data >> From Database menu. Furthermore, the Load Data >> From computer alternative gives different data handling capabilities such as interpolate the data, set the number of clusters, set the number of environmental variables, eliminate or add rows, crop the data, set the data for simulation or forecasting.

The “**Case Selection**” tab gives the alternative to do simulation or forecasting. These process can not be performed until the parameters are well configured in the “**Parameters**” tab. The Parameters can be set in the current value or can be predicted in function of the data already loaded in the system. Figures 4, 5, 6, and 7 shows the simulation, forecasting, parameter current value, and parameters prediction tabs. The Map tabs open the opportunity to visualize the data obtained from the database. There is no alternative to load the data manually since the visualization requires to set the latitude and longitude geo-referencing information of each cluster under consideration. Figure 8 shows the visualization of this tab. Finally, the Clear tab allows to eliminate the data saved and clear the figures while the information/help menu shows information regarding the parameters used in the models and the information regarding the software. The following sections of this manual will describe the functionalities involved in the main tabs and how to use them. First the different load options will be analysed. Then the parameters configuration. And finally, simulation, forecasting, and map interaction will be described.

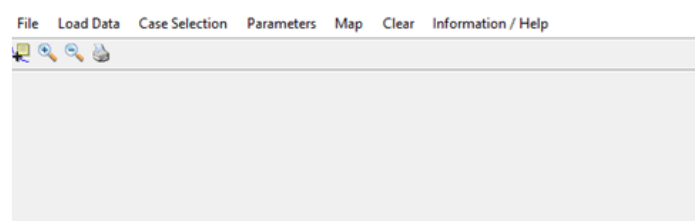


Figure 1. Introductory Menu

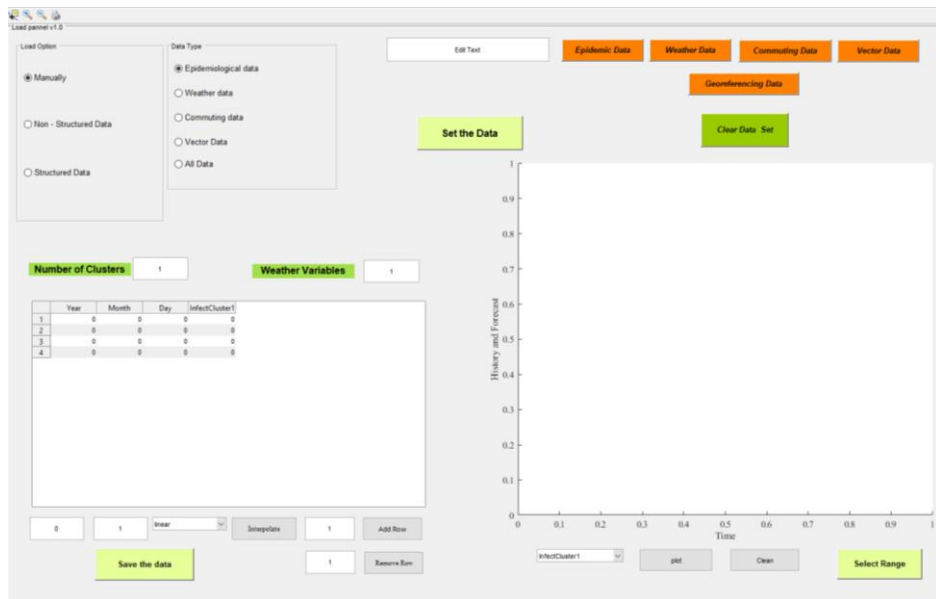


Figure 2. Load Data >> From Computer

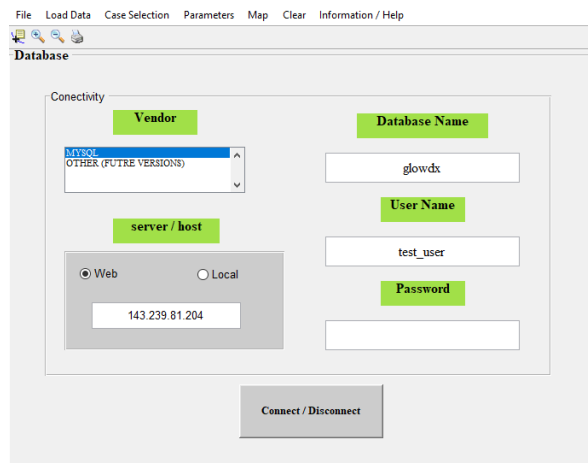


Figure 3. Load data >> From database

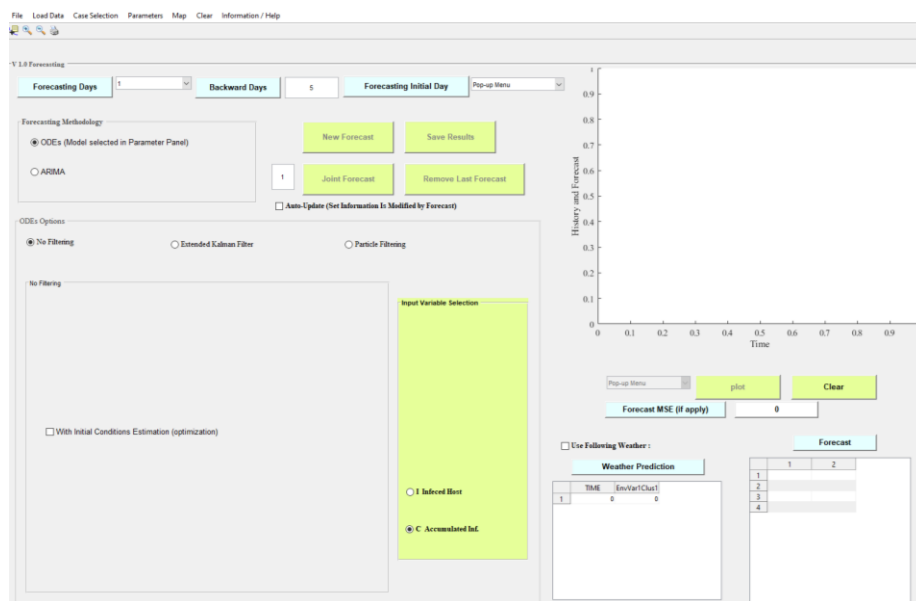


Figure 4. Case Selection >> Forecasting

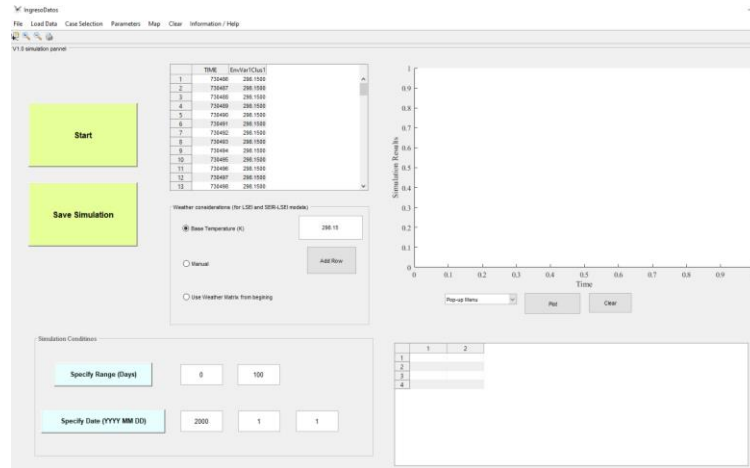


Figure 5. Case Selection >> Simulation

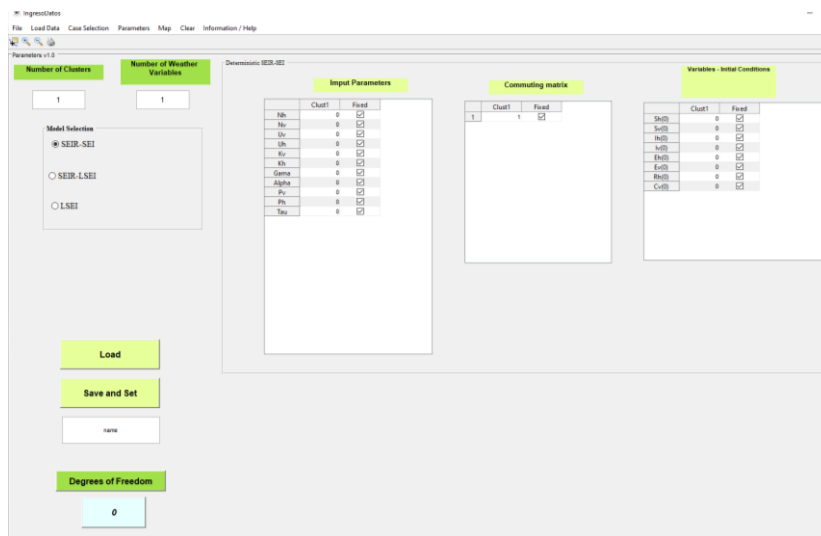


Figure 6. Parameters >> Current Values

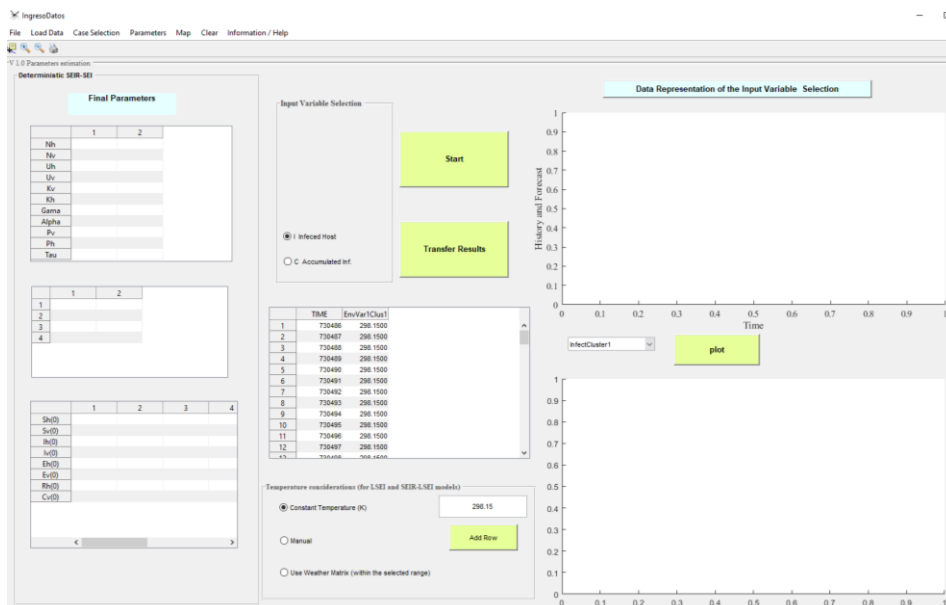


Figure 7. Parameters >> Prediction.

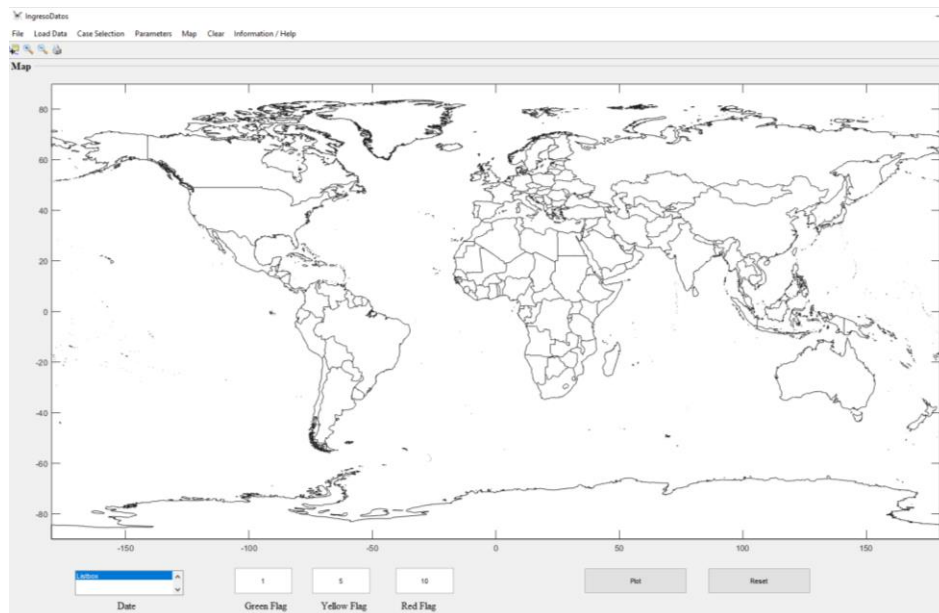


Figure 8. Map

- **Load Options**

Load tab: As observed in the Load Data tab, there are three load options. Manually, Non-structured, and Structured Data. The manually load option allows to interact with the table and set the data for simulation and forecasting. Furthermore, the manually configured load option allows to incorporate the data in timestamps as Year, Month, and Day style. As you choose the different options of load, the table configuration changes depending on the data style and the data type (explained in the next paragraphs).

Independent of the style of load option, new rows can be added by the “Add Row” button on a specified line (square at the side button). Care must be taken at all times that days are not repeated and that the information follows sound formatting. Independent of the load option, the type of data to be incorporated in the system has to be of four different styles. It can be Epidemiological data, Weather data, Commuting data, or Vector Data. The following examples show how to load non-structured data and Structured data (It will be considered two files included ***Iquitos_Training_Data.xlsx*** and ***IquitosT_Training_Data_Eduardo.xlsx***).

Load Example 1. First select Structured Data and Epidemiological data. After, change the number of clusters to two. Then press the “Select File” button and select the corresponding file (***Iquitos_Training_Data.xlsx***). A popup menu should show up specifying that the number of columns in the file is bigger than the number of clusters set. This happens since there are several columns in the file attached and the program identifies always in the structured data that the first column is the time stamp, and the following correspond to the data of a given data type. Press yes and accept to use only the first two columns of data. By doing this the table will show up the loaded data and the figure will show up the data historical behaviour as represented in the following figure. The data have not been set yet. The process of setting is to tell that the data described in the table is the one that will be used for parameters identification, simulation and/or forecasting. To set a data you have to press the “Set the Data” button. Once you press that button, depending on the data type selected, the orange squares will turn into a blue colour instead of orange, describing that that type of data is ready to be used by the software. The data can be handled in different ways.

First, as can be observed, the time is represented by a number. This number represents the number of days passed from the year 0, month 0, and day 0 of the loaded data. Once loaded, two squares

describe the maximum and minimum days loaded. (should be 730668 and 733979). You can remove a line (e.g. remove the line 11 by replacing the value 1 by 11 and pressing the “Remove row” button). As observed that point has been removed. Furthermore, rows can be added by selecting the position it will be incorporated and the use of the “Add Row Button”. Finally the data can be interpolated using *linear interpolation*, *spline*, and *pchip* methodologies (refer to Mathworks information for further understanding of these methodologies). Select for example spline and press the interpolate. As observed the interpolation process results (Figure 10) Interpolation is performed between the range set within the edit boxes. At any time you can modify the range of interpolation but once the process has been performed, the data have to be loaded again or modify the interpolation technique. Finally, if it is desired to work with only a range of the data, under the tab menus, there are four figures. They correspond to data graph visualization, zoom in, zoom out, and print the current screen. One can use the zoom in to specify a range and the “Select Range” button under the figure to eliminate all the data outside the selected range.

Under the figure can be also find the data show drop menu that allows selecting the most relevant data information to be displayed in the figure. Select what data to plot and press the “plot” button.

Load Example 2. As for a second alternative select non-structured data and weather data type. Select the ***IquitosT_Training_Data_Eduardo.CSV*** file and load it considering only 1 cluster (CSV files are handled with only 1 cluster in considerations use excel file or database information to handle more than one at a time). As you specify the data, the system requires you to specify the Columns by letters / numbers in which the year, month, and days are found in the file. Later on specify the range for the corresponding data (1,2,3,4) in this case. Avoid to load CSV files this way since if a CSV will be loaded, only one column can be used for data. In the case of Excel files are handled this way, the non-structured data will allow to load range of data. Care must also be taken in function of the version of the excel file saved, since depending on the version and language used, the CSV files can be interpreted differently (i.e. avoid the use of CSV files in general). After loading the data remember to press the “Set the Data” Button to load it. If the commuting data is loaded and set, this will be sent directly to the parameters and not kept in memory for further uses.

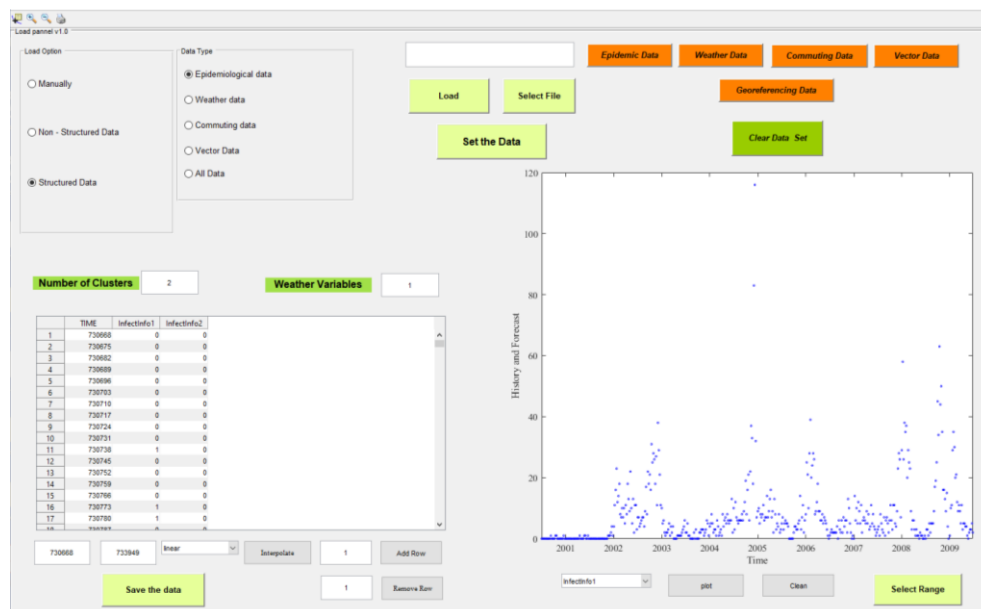


Figure 9. Structured Data loaded.

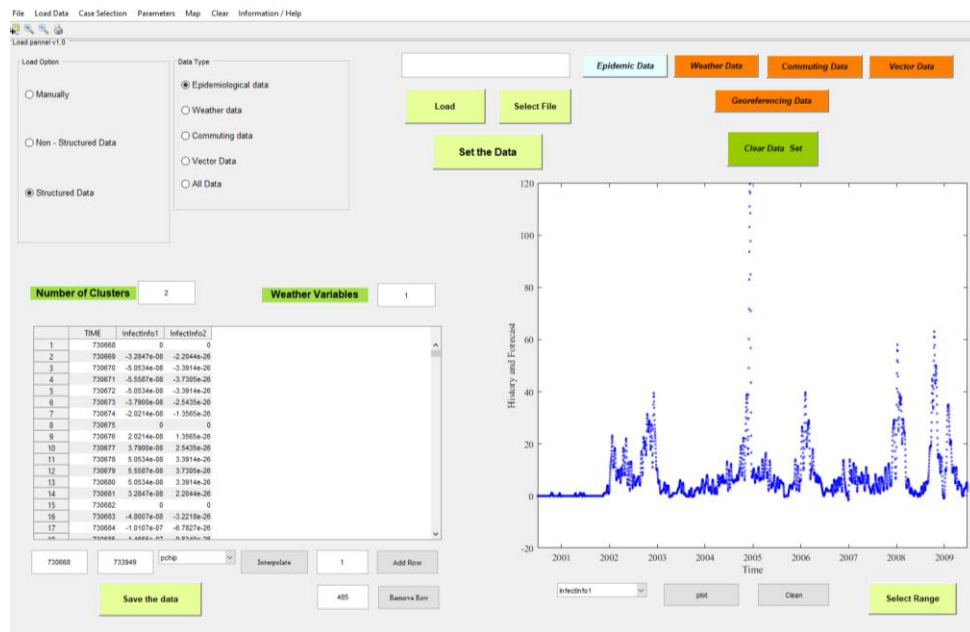


Figure 10. Interpolated data

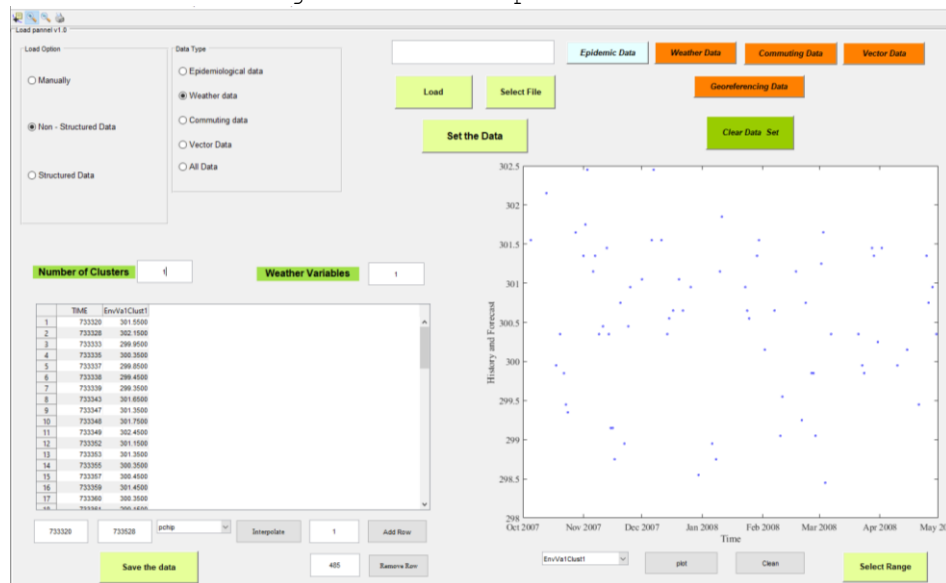


Figure 11. CSV load with non-structured data.

Database tab: In the case of the data being loaded from the databases, first, a direct connection to it has to be created. Figure 12 shows the first screen opened to interact with databases. The current version of the software only interacts with MYSQL databases. The database can be local or web, but a clear connectivity information is required to handle queries. The Database handling could require some further toolboxes from matlab in order to run properly (depending on the current installation existing in your computer). The current version of Matlab used for SQL connectivity is Matlab R2019a. Some connectivity issues could come if previous versions are used. As observed in Figure 12, the information of Database Name, User Name, Password, and server / host has to be incorporated in order to connect with the database. Considering the current setting, press the “Connect/Disconnect” button. Once the connections happen properly, further windows will pop up (Figure 13). The new windows correspond to Info Loader, Query, and Panel interactions. Once connected to the database, the Info loader windows shows the tables that exist in the given catalog. You can press the load sample

to shows some of the information of the given table. Figure 14 shows a representation of the GLOWDX table that contains several data information.

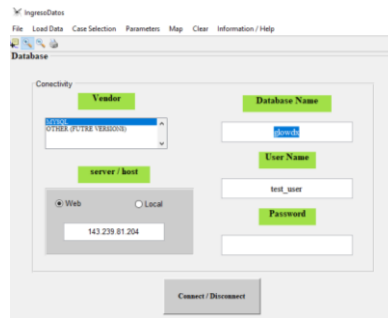


Figure 12. Database load menu.

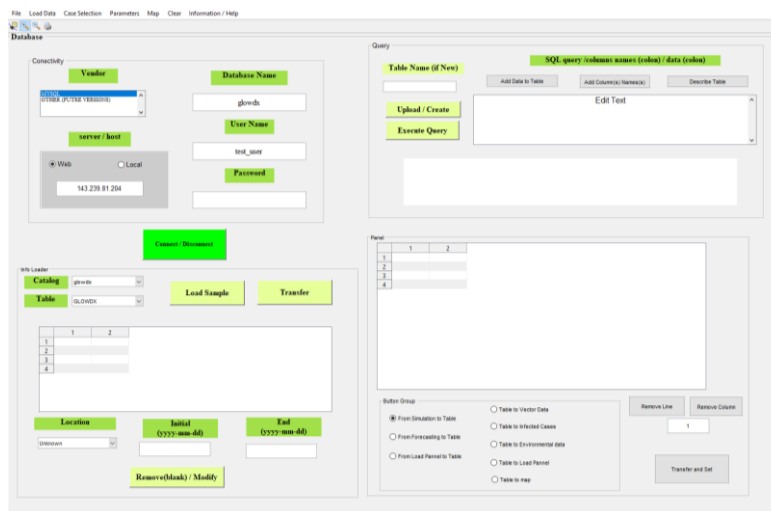


Figure 13. Extended Database Panel.

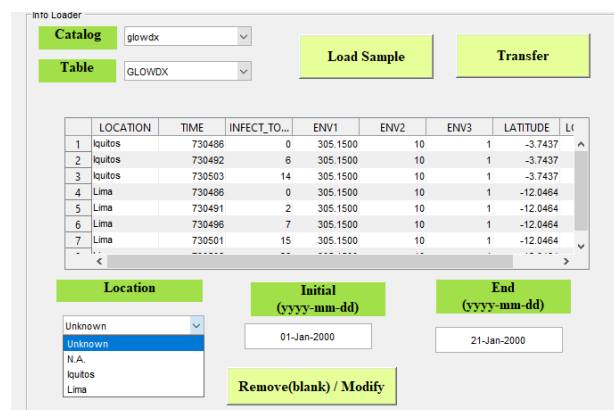


Figure 14. Glowdx Database data.

The GLOWDX table contains the SQL and software reserved words LOCATION, TIME, INFECT_TOTAL, ENV1, ENV2, ENV3, LATITUDE, LONGITUDE. If you want to create a table in the database and make the software to recognize this data as given its nature (e.g. environmental variable) you have to make sure that the table includes the reserved words. LOCATION corresponds to a given name that will be correlated to a unique combination with the latitude and longitude of the specified region. TIME shows the day past using as referencing the 0 year, 0 month, and 0 day information. Infect_Total gives

the number of infected people from a vector-born disease for the given day and location. ENV1 up to ENV3 are the environmental variables to be considered in the location at a given day. Finally, LATITUDE and LONGITUD are the geo-referencing information for map interaction. Furthermore as observed in Figure 14, once a sample is loaded the Location drop down menu describe the alternatives for transfer specific locations, Unknown location (i.e. save all unknown locations with this name), and the N.A. which represent all the locations at the same time.

The initial and end dates obtained from the table are reported in the same windows. To transfer a specific location information to the processing windows, use the “Transfer” button and choose the specific location. As an example, only the Iquitos location were sent to the corresponding windows (Figure 15). You can also select N.A. as a region to transfer all the information to the corresponding handling windows. The transfer Panel will load the whole data from the location and will show the information. You can remove a specific line or column from the table using the Remove line or Remove Column button. The most important feature of this windows is to transfer the data from the current table state to the Vector Data, Infected Case Information, Environmental data, to the map representation, or to the Load Panel. Furthermore, results from the simulation, forecasting, and load panel can be transferred to this table in order to save it in the database. Once you decide what you would like to perform press the “Transfer and Set” button.

	LOCATION	TIME	INFECT_TO...	ENV1	ENV2	ENV3	LATITUDE	LONGITUDE
1	Iquitos	730406	0	305.1500	10	1	-3.7437	-73.2516
2	Iquitos	730402	6	305.1500	10	1	-3.7437	-73.2516
3	Iquitos	730503	14	305.1500	10	1	-3.7437	-73.2516

Button Group

☒ From Simulation to Table
☐ From Forecasting to Table
☐ From Load Panel to Table

☐ Table to Vector Data
☐ Table to Infected Cases
☐ Table to Environmental data
☐ Table to Load Panel
☐ Table to map

Remove Line Remove Column

1

Transfer and Set

Figure 15. Transfer window.

Figure 16 shows the results of transferring the loaded data (1 cluster) to the current table in the Transfer window. As observed, Environmental variables that are not specified are not included and will be required to be modified properly.

	LOCATION	TIME	INFECT_TO...	ENV1	ENV2	ENV3	LATITUDE	LONGITUD
1	Iquitos	2005-07-01	0					
2	Iquitos	2005-07-02	0					
3	Iquitos	2005-07-03	0					
4	Iquitos	2005-07-04	0					
5	Iquitos	2005-07-05	0					
6	Iquitos	2005-07-06	0					
7	Iquitos	2005-07-07	0					
8	Iquitos	2005-07-08	0					
9	Iquitos	2005-07-09	0					
10	Iquitos	2005-07-10	0					
11	Iquitos	2005-07-11	0					
12	Iquitos	2005-07-12	0					
13	Iquitos	2005-07-13	0					
14	Iquitos	2005-07-14	0					

Button Group

☐ From Simulation to Table
☐ From Forecasting to Table
☒ From Load Panel to Table

☐ Table to Vector Data
☐ Table to Infected Cases
☐ Table to Environmental data
☐ Table to Load Panel
☐ Table to map

Remove Line Remove Column

1

Transfer and Set

Figure 16. Transfer windows after transfer from the load panel.

Finally, The Query windows (Figure 17) allows to:

- Execute queries and process them (by using SQL queries)
- Create a new table in the database

- Update an existent table in the database
- Add data to table (by using colon separated information)
- Add names or modify names of tables for database modifications.

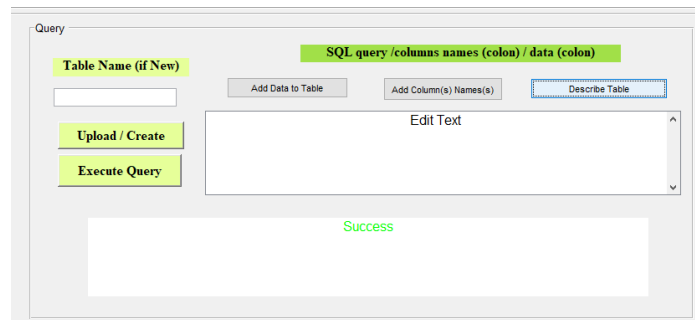


Figure 17. Query Windows

In order to create a new table, write a non-existent name in the table name space and then press the upload/create button. To upload an existent table, press the upload/create button. Once you write an SQL query in the Edit Text area, press the Execute Query for its processing. Since the queries are database dependant, once you refer to a given Table, make sure to use the *catalog.tablename* referencing. The database connection use a JDBC (Java database controller) make sure to install it as a static file. If not set properly, the software will require the file and set it as a dynamic object (will have to be set each time the software is opened). It is recommended the installation of the following version of the JDBC connector: **mysql-connector-java-5.1.47-bin.jar** that can be obtained directly from the mysql website.

- **Parameters**

Parameters>>current values: Figure 18 shows the Parameters>>current value panel (with the SEIR-LSEI_2C2E file loaded; explained later). In this panel the number of clusters and weather variables can also be modified and changed. In this panel all the configurations are sent to the load panel (and vice versa). The first decision to perform regarding the parameter panel is the selection of the model that will be implemented. It can be the classical SEIR-SEI model, The pure vector model (which do not work with any cluster), and the combination of the SEIR-SEI model with the vector model (i.e. the complete SEIR-LSEI model). As you change the model, the parameters to be fix changes to. To load present configuration parameters, press the “Load” button and choose a “.mat” file with the corresponding data. There have been included in the software the following files in order for the user to interact with pre-configured data.

- Iquitos1 (Parameters preconfigured for the Iquitos data using the SEIR-SEI model)
- Iquitos2 (Parameters preconfigured for the Iquitos data using the SEIR-SEI model with reinfection)
- Iquitos3 (Parameters using the SEIR-LSEI model for the Iquitos data-base)
- Iquitos4 (Parameters using the SEIR-LSEI mode for the Iquitos data-base with full selection option)
- SEIR_LSEI_1C1E; SEIR_LSEI_2C2E;
- SEIR_SEI_1C1E
- LSEI_2C2E.

File Load Data Case Selection Parameters Param Map Clear Infores / Help

Parameters v1.0

Number of Clusters

2

Number of Weather Variables

1

Deimos HSB-LSEI

Model Selection

☐ SEIR-SEI
☒ SEIR-LSEI
☐ LSEI

Input Parameters

	Cust?	Fwd?	Cust?
NH	2110051	☐	2110051
UN	3405004-05	☐	3405004-05
KC	0.1250	☐	0.1250
SH	0.2500	☐	0.2500
Geme	0.1007	☐	0.1007
Alpha	0.4517	☐	0.4517
Psi	0.2379	☐	0.2379
Phi	0.2190	☐	0.2190
Tau	0.0202	☐	0.0202
In	210004-04	☐	210004-04
ep	30	☐	30
Tp	1	☐	1
Tt	1	☐	1
Ts	1	☐	1
Tp	1	☐	1
Prw01	0.4000	☐	0.4000
Prw02	0.0007	☐	0.0007
Prw03	1	☐	1
Mu_1	200.1500	☐	200.1500
Mu_1	200.1500	☐	200.1500
Mu_1	200.1500	☐	200.1500
Mu_1	200.1500	☐	200.1500
Mu_1	200.1500	☐	200.1500
Mu_1	200.1500	☐	200.1500
Mu_1	200.1500	☐	200.1500
Mu_1	200.1500	☐	200.1500

Commuting matrix

	Fwd?	Cust?	Fwd?
1	1	☐	0
2	300	☐	0.0000

Variables initial conditions (then applicable for predictions)

	Cust?	Fwd?	Cust?
Sin(0)	2110001	☐	2110001
Sun(0)	430700	☐	430700
Rin(0)	1000	☐	0
Iin(0)	1200	☐	1200
Ein(0)	253.2040	☐	253.2040
Rin(0)	951.0000	☐	951.0000
Rin(0)	211.0015	☐	211.0015
Cin(0)	0	☐	0
Ein(0)	10	☐	10
Iin(0)	10	☐	10
Rin(0)	10	☐	10
Sin(0)	10000	☐	10000

Load

Save and Set

DESI_LSEI_NC_mst

Degrees of Freedom

0

Figure 18. Parameter Current Value Panel

As observed in the tables, each parameter has been assigned a value and a fix column at its side. These checks are used during the different optimizations performed in the program (for parameter determination and forecasting) therefore it has to be decided before performing these processes which variables/parameters will normally be fixed and which variables/parameters will be determinate. As the number of parameters is modified, the Degrees of Freedom describe how many variables are predicted. As the number of variables/parameters to estimate growth, the optimizations are cumbersome and could lead to extensive prediction times. So care must be taken in order to estimate several parameters. It is recommended to leave the models state variables non-fixed with some parameters to be determinate.

Simulation: The simulation panel is shown in Figure 19. As observed in the figure, the simulation have the alternatives of specify different environmental variables by manual configuration, homogeneous configuration (i.e. repeated between each cluster), and use of the loaded Weather Information. If you decide to put two different environmental variables, you can place the variables values separated by space in the text box and the software will set the environmental variables properly. There are two green buttons, one that start the simulation using the configuration stablished in the Parameters panel, and the other that save the results of simulation. The save simulation results alternative will save all the variables, while if you want to select the most relevant variable depending on the model, use the File tab at the top of the windows. Once you start the simulation, there is going to use the parameters set in the parameter panel as initial states. In the “Specify range days” you can set the initial and final number of days in the simulation. This assumes that a consecutive number of days will be used (i.e. not in a weekly base). Furthermore, you can specify the calendar initial day of the simulation. Once the simulation is performed, the results will be displayed in the graph and in the table, as shown in Figure 19. The drop menu can be selected (with the plot button) to plot a specific variable.

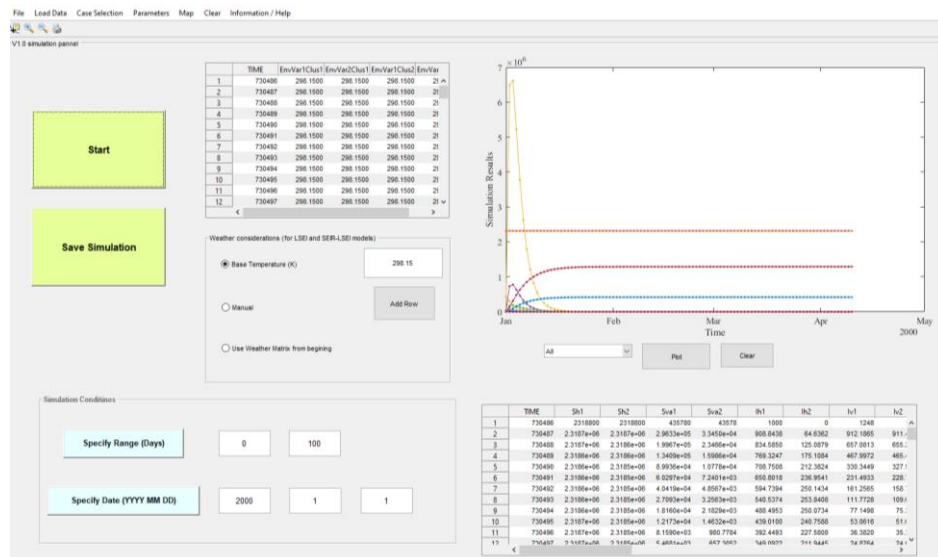


Figure 19. Simulation Panel.

Parameters>>determinate: To perform the parameter determination, let's assume that you will use the data from the Iquitos between the day 73365 and 733512. This data have been set (without the environmental variables yet). Furthermore, use the Iquitos1 parameters (in the parameters current value and load them). First, make sure that you will uncheck all the state variables and leave only Nh and Nv as check variables (as shown in the following figure). Since Iquitos1 correspond to SEIR-SEI model, it automatically set the system configuration to 1 cluster and the corresponding model, nevertheless the number of weather variables (not important for this model) have to be set always by the user. The Parameter Estimation panel should be similar to Figure 20. As observed in the figure, the data representation of the set information can also be visualized. This panel has different windows. At the top left, are the parameters that were obtained after the prediction. You can transfer these results (buy the "Transfer Results" button) to the Parameter>>Current Values panel. The Input Variable Selection window describes the type of data that will be used for performing the prediction. For the SEIR-SEI you can only use the Infected Host as predictor or the Number of Accumulated case as predictor. Under this window you have the environmental variables information that will be used. In this case, similar as explained before, you can set constant values (separated by space) manual incorporated values, and use the set information from the load panel.

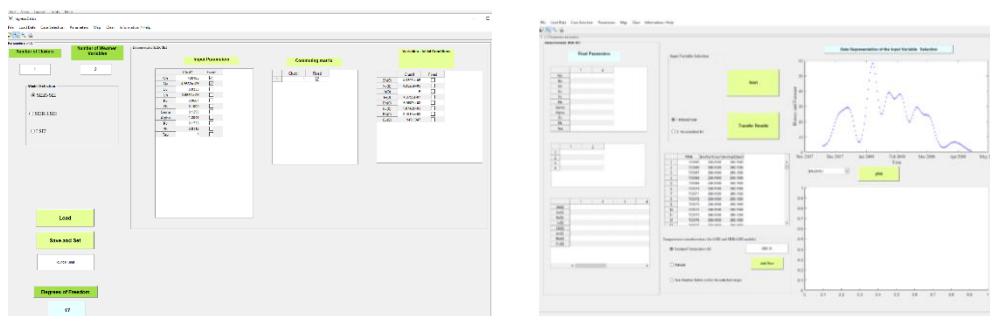


Figure 20. Parameter Estimation Panel with the set data.

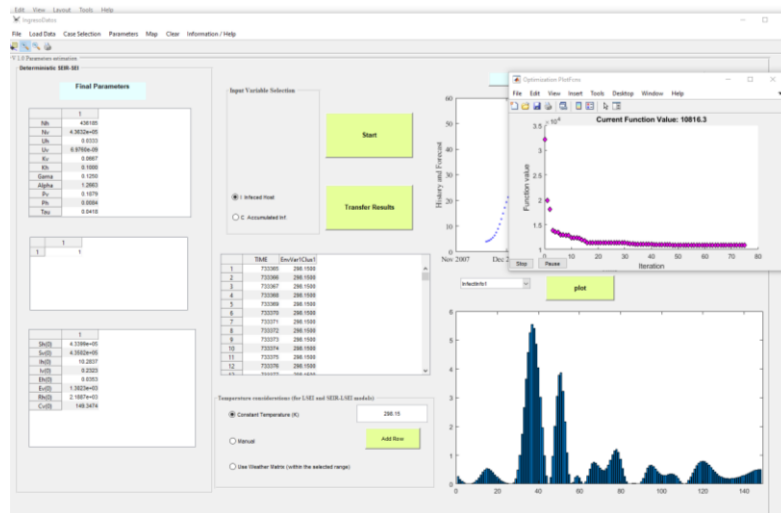


Figure 21. Parameter Estimation Results.

Once the system is configured, a new graph will pop up showing the error committed by the objective function (simulated values vs submitted values) as the optimization continues. Once the process ends, the bottom windows will show the simulation results as deviation with respect to the loaded data. The left tables will show the final parameters in function of the model selected.

- **Forecasting**

Once the data is loaded in the system (load panel) you have full access to the forecasting options. The forecasting windows is constituted by four different regions. The forecasting methodology window, the solver configuration (windows under the Forecasting Methodology Window). The results section, and the pre-setting section (top buttons, including New Forecast, Joint Forecast, Save Results, and Remove Last Forecast Buttons). The first decision to make in the forecasting is to choose between timeseries forecasting methodologies (ARIMA and SARIMA) and Ordinary Differential Equations (ODEs) based forecast. If ARIMAS are considered to be used in the software with matlab, the proper toolbox have to be included in the Matlab version installed in your computer. Alternatively, you can eliminate all the components in the IngresoDatos Files that are related to ARIMA (i.e. `function pushbutton124_Callback; radiobARIMA_Callback; pushbutton122_Callback`). The included Model ARIMA is based on the TSAFT guide that is also provided with the software. You can open it and use it independently. The ODEs based software are not related to any Matlab toolbox so it can operate independently.

At the top of the Forecasting panel, three settings have to be made before making the forecasting. The Forecasting Days drop down menu specify the number of days that will be forecasted. The Backward Days specify the number of points, including the initial day, that will be used for parameter determination. Finally, the Forecasting Initial Day specify the starting point for the forecasting. The drop down menu will show only existing days.

ODEs Forecasting: If ODEs are going to be used to do the forecasting, the Model that will be used is depending on the setting established in the Parameters>>current value menu. The type o variable that is used is set in the Input Variable Selection windows that is inside the ODEs option windows. There are three general family of forecasting methodologies, independent of the model used for the forecasting. There are those based on pure ODEs, those that use the Extended Kalman Filter to improve forecasting, and those that use the Particle Filtering. Depending on these selections, further configuration will be possible. Figure 22 shows the different alternatives.

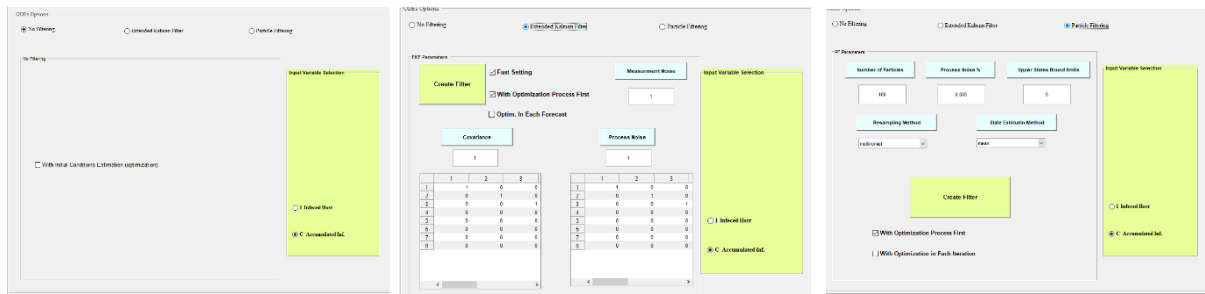


Figure 22. Different ODE options.

The No filtering perform the forecasting by using the parameters from the parameters>>current Value windows. These parameters can or not be optimized depending if the check box “With Initial Condition Estimation” is marked. Figure 23 shows the Forecasting results obtained without the optimization. As observed, a Mean Square Error, if possible to be estimated, is calculated based (see the square button results in Forecast MSE). The table presents in the first column the days considered (5 forecast, 5 backward, 1 present day) and the state variables in the similar order than those presented in the parameter estimation windows.

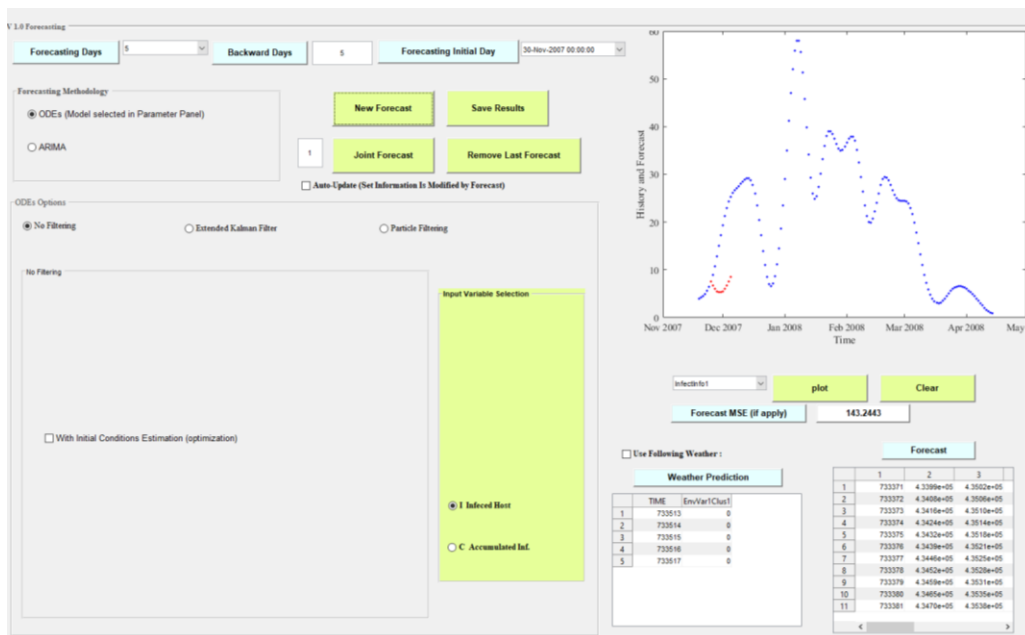


Figure 23. Forecasting with ODE systems and without Optimization

To allow optimization within the forecasting, the checkbox “With Initial Conditions Estimation” Has to be checked. As shown in the Figure 24. The results are considerably improved (from a MSE of 143.243 to 0.55027) if optimization is performed. As each forecasting is performed, the new results are displayed in the same graphs. The “Remove Last Forecast” button can be used to delete from memory and the graph the last forecast.

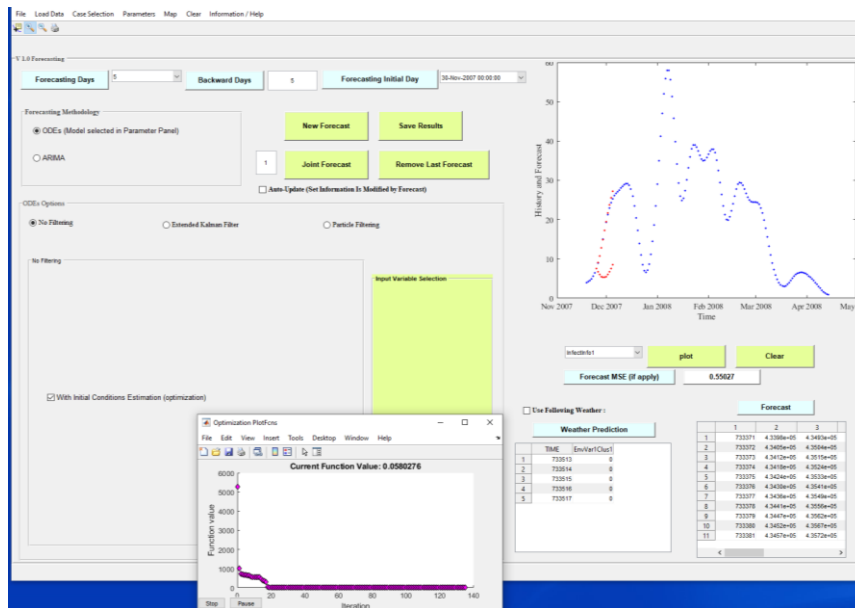


Figure 24. Forecast with Optimization.

The “Joint Forecast” button allows to perform several forecasting one after another using or not optimization. Figure 25 shows the results obtained by using the SEIR-SEI model in which 15 consecutive forecasting is performed. The software by itself jumps the forecasting for one day into another. The optimization processes performed in each forecast within the series of forecast will be restricted to obtain the values, in each iteration, of the variables left unset in the parameters>>current values menu.

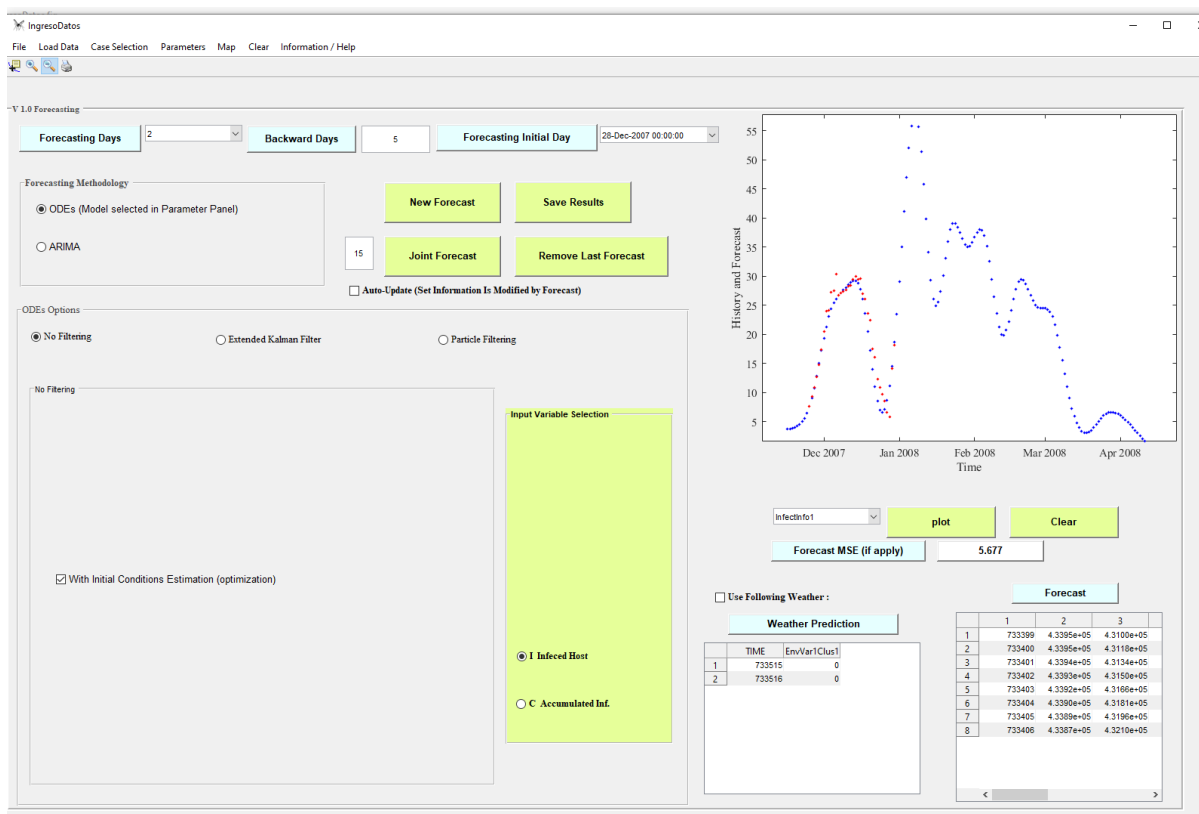


Figure 25. Joint Forecasting.

If a forecast with Extended Kalman Filter is desired to be done, first, the filter has to be created. To do this first decide the methodology to create the filter by using the predefined checkboxes. The Fast Setting allows to set the covariance and process noise as diagonal values of the values fixed in the Covariance and Process Noise boxes. “With Optimization Process First” allows to perform an optimization to set parameters before the final filter is created. The “Optimization In Each Forecast” box allows to perform several optimizations between forecasts when the “Joint Forecast” is used. The number of consecutive days is fixed by the number set at the side of the “Joint Forecast” button.

Once you press the Create Filter Button, and if the optimization (or not) process is finished, a warning dialog will popup telling you that the filter is ready to be used. Once the filter is ready, you can perform the next Forecasting. By pressing the “New Forecast Button”

Theory regarding “Extended Kalman Filter” is not the current objective of this manual.

Similarly to the Extended Kalman Filter, you can use the Particle filter by first creating the corresponding filter. In this case, the parameters that have to be set are the Number of Particles to be considered, the process noise percentage, the resampling method, the state estimation method, and the upper state bound limits (Covariance of each values since it is considered a diagonal matrix). Care must be taken in the number of particles used since as the number increases, there is extensive periods of time required to perform the forecasting.

Finally, The ARIMA forecasting is a completely different methodology of forecasting that is purely based in data and do not require time as variable. We will be using to file name “test” as an example to use it. If you will be load this file since it do not possess any time stamp recognizable by the system to do any plot so a message mentioning this plotting problems will show up. After the data is loaded, you can go to the ARIMA selection option in the Forecasting tab. Figure 26 shows the how the data looks up.

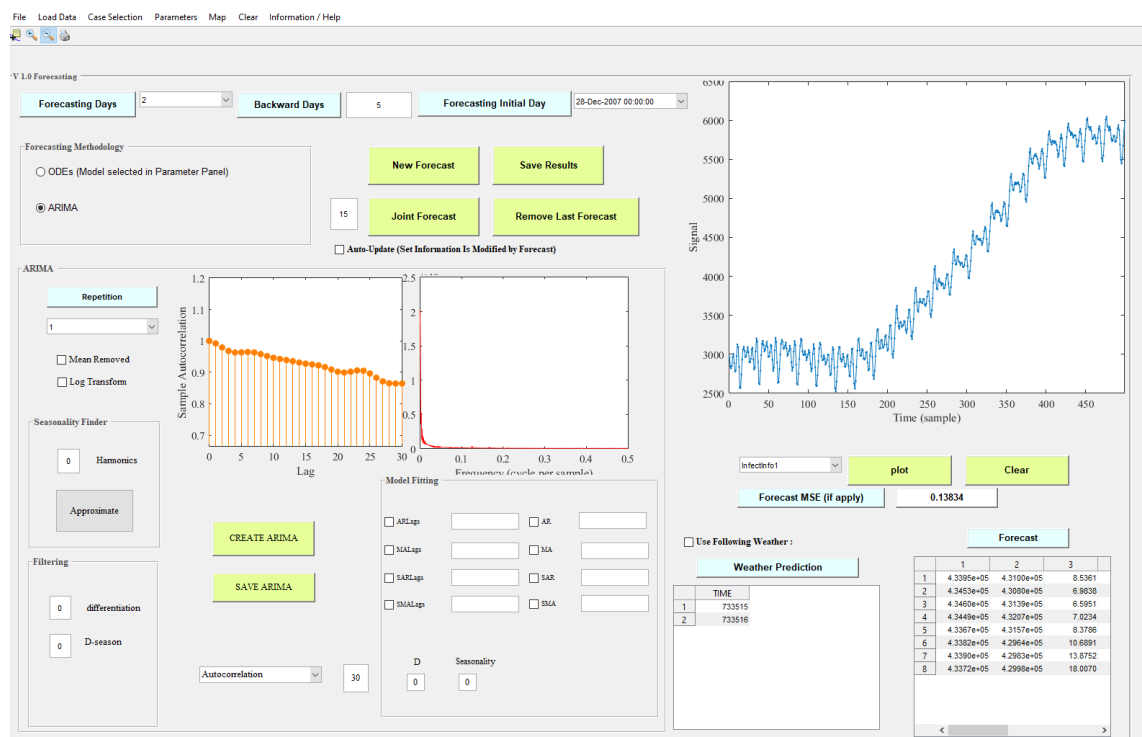


Figure 26. ARIMA forecasting.

This data have to be handled in order to be useful for prediction. The checkbox “Mean Removed” and “Log Transform” can directly be used for remove average values of apply a log transformation to the data loaded. In this case, there is no required any transformation since the data do not show an exponential behaviour. Since the data is continually increasing, the data will be differentiated. To do this apply a differentiation of order 1. Press enter and the data will be transformed accordingly. Secondly, since a periodicity is observed in the Frequency domain graph, you have to apply a d-seasonality to the data. To do this choose 3 harmonics and press the “Approximate” button. Figure 27 a shows the results obtained by doing the previous transformations. Once you place a value in the d-season square equal to 24, you can estimate the Autoregressive Moving Average lags to be used. By switching between the Partial Autocorrelation and Autocorrelation graphs in the dropdown menu, you will be able to estimate these values. (Theory behind this is left for the users and is not included here). In this case, The Auto Regressive lags (AR) are set to 2 (1 2 in the ARLags edit box). Press the Create ARIMA to estimate the ARIMA parameters and the process is ready to do the forecast. Figure 28 shows the results using the ARIMA alternative.



Figure 27. ARIMA Transformation.

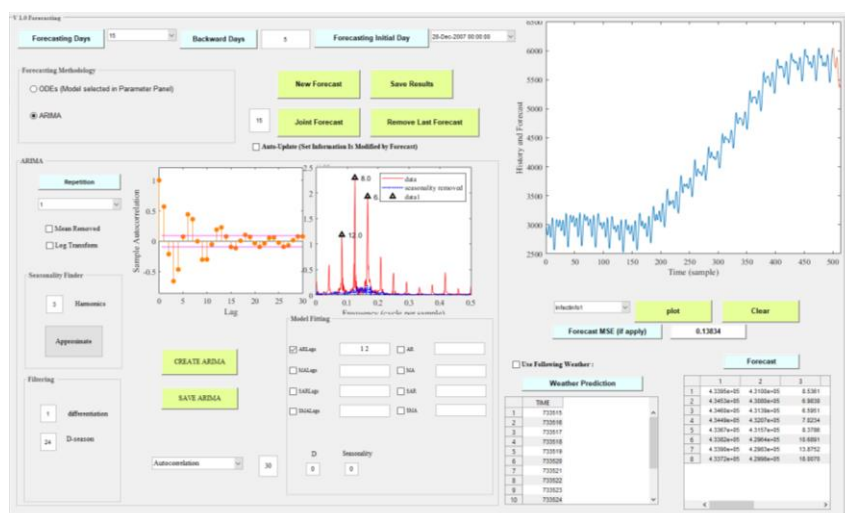


Figure 28. ARIMA forecasting Results.

- Mapping.

The mapping tab can be used to graphically display the information saved in the database. To do this first, you have to transfer from the database the information of a given region or all the regions saved in the database. To do this, in the Load Data Panel>>Database, you have to choose the alternative “Table to Map” and press the “Transfer and Set” button. After you can go to the Map tab. Once the data is correctly transfer you can specify a date in the Date scroll down box and press the button “Plot” to visualize how the diseases are behaving in function of the data loaded with respect to the geographical information set. The “Green”, “Yellow”, and “Red” flags are used to set limits in the number of reported cases and graphically place colours on the data observed. Figure 29 shows the results obtained once for the 18-January-2000 based on an information loaded from the database.

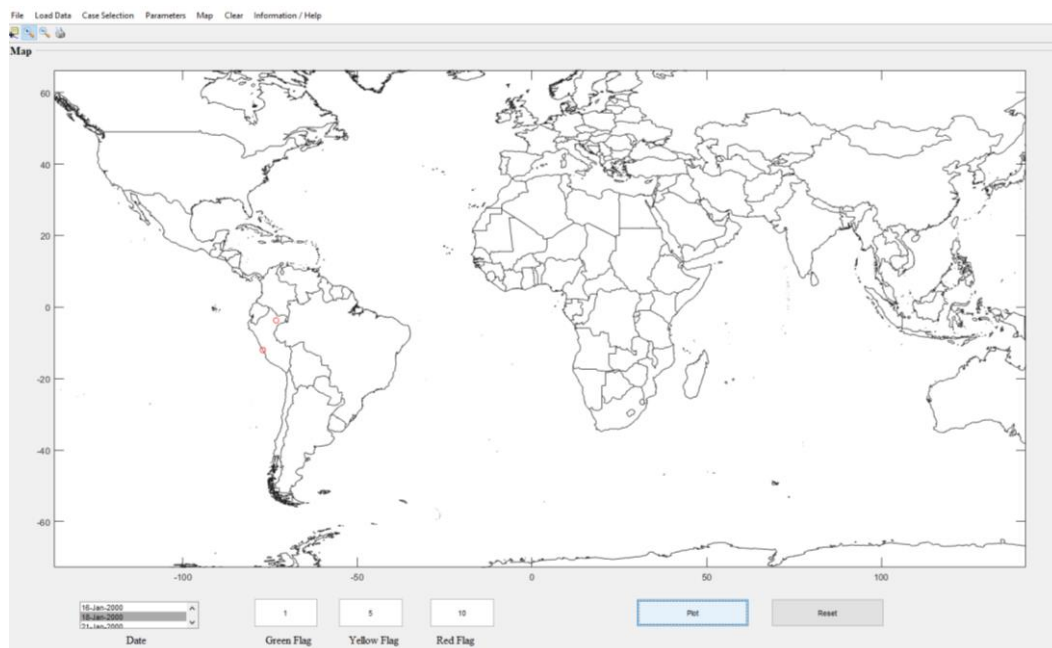


Figure 29. Map Results

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